

Ajaykumar K

AI & ML Engineer | Computer Science & Engineering

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PROFESSIONAL SUMMARY

First-year B.E. student specializing in AI & ML with hands-on experience building intelligent systems — from CNN-based crop disease classifiers to real-time voice assistants deployed on mobile. Skilled in Python, TensorFlow, OpenCV, NLP, and vector databases (ChromaDB, Pinecone). Actively engaged in research, having presented papers at multiple institutions. Seeking opportunities to apply AI expertise to real-world, high-impact problems.

EDUCATION

B.E. in Computer Science & Engineering (AI & ML)

2025 – 2029

K.S. Rangasamy College of Technology, Namakkal

SGPA: 8.71 / 10

SKILLS & INTERESTS

Programming

Python, C

AI / ML

TensorFlow, OpenCV, NLP

Web Development

HTML, CSS, JavaScript

Databases

MySQL, ChromaDB, Pinecone (Vector DBs)

Tools

Git, GitHub, Unity, LiveKit

Interests

Robotics, Web Development, Sustainable Farming, Chatbots

PROJECTS

Smart Farming — AI Crop Disease Detection

2026

- Built a CNN-based crop disease classifier using **TensorFlow and OpenCV**, enabling farmers to identify diseases from crop images in real time.
- Designed an end-to-end pipeline — image capture, preprocessing, model inference, and result display — reducing disease identification time significantly.
- Presented as a research paper at Sri Eshwar College of Technology; model optimized for low-connectivity rural environments.

Voice Assistant — Real-Time Mobile Voice Agent

2026

- Developed a voice assistant that interprets user commands and guides task completion through natural voice interaction.
- Deployed on mobile using **LiveKit** for real-time, low-latency voice streaming with improved user experience.
- Integrated **NLP** modules for intent recognition, enabling context-aware responses across multiple use cases.

ACTIVITIES & RESEARCH

- Participated in the **AI in Water Resources** workshop at Vellore Institute of Technology — explored AI-driven solutions for resource management.
- Presented research paper on **Smart Farming** at Sri Eshwar College of Technology — demonstrated CNN-based disease detection system.
- Presented research paper on **AI Hallucination** at Coimbatore Institute of Technology — analyzed failure modes in large language models.

CERTIFICATIONS

- **Data Analytics Job Simulation** — Deloitte (via Forage)
- **Python Essentials** — Netacad (Cisco)
- **Artificial Intelligence** — IBM
- **C Programming** — Infosys Springboard
- **Digital Edge** — NASSCOM